

Pakistan Tech Compensation Audit: Does a Pay Gap Persist After Controlling for Experience?

Author: Chander **Data source:** 2025 Stack Overflow Developer Survey (public release) **Analysis period:** July 2026

1. Business Problem

Compensation disparities in tech are frequently discussed but rarely tested rigorously. Most informal comparisons stop at "Group A earns less than Group B" without asking the question that actually matters to HR leaders, policymakers, and affected developers themselves: ****does the gap survive once you control for the legitimate drivers of pay, like experience?****

This project was originally scoped as a gender pay gap analysis. After downloading the 2025 survey data, no gender field was present in the public release (Stack Overflow appears to have removed optional demographic questions in recent survey years). The project was reframed around a comparison with direct personal relevance: ****Pakistan-based developers vs. the rest of the world****, using the same statistical framework originally planned for the gender comparison.

Business question: Do Pakistani developers earn less than global peers, and if so, does that gap persist after controlling for experience — or is it explained away by legitimate factors?

Population of inference: Respondents to the 2025 Stack Overflow Developer Survey. This is a self-selected, internet-active, largely English-fluent population — not a representative sample of the Pakistani or global tech workforce. All findings below are scoped accordingly.

2. Data & Methodology

2.1 Source and Sample

The raw dataset contains 48,882 respondents across ~173 fields. After filtering to respondents with non-null `Country`, `ConvertedCompYearly` (annual compensation, USD-converted), and `YearsCode`, the working sample was 23,838 rows.

2.2 Cleaning Decisions and Reasoning

Decision	Reasoning
Salary bounds: \$1,000–\$2,000,000	Global data included implausible values (min \$1, max \$50,000,000). Rather than a mechanical statistical rule (e.g., IQR), which performs poorly on heavily right-skewed income data, bounds were set using domain judgment: no full-time developer role plausibly pays below \$1,000/year or above \$2,000,000/year.
Investigated Pakistan-specific impact of the floor	Pakistan lost 20/132 rows (15.2%) to the floor vs. 3.05% overall — a 5x higher removal rate, worth investigating rather than accepting silently. Inspection of the excluded values showed several exact duplicate entries (e.g., five respondents all reporting \$351), indicating a likely systematic unit-conversion error rather than genuine low salaries. This supported keeping the floor as-is.
YearsCode capped at ≤50	Values above 50 (up to 100) were concentrated in round/repeated numbers consistent with the survey's "more than 50 years" option being converted to a sentinel value rather than a true reported figure. Affected 0.33% of rows, entirely within Rest-of-World — no impact on the Pakistan subgroup.

Final sample: 23,036 respondents — 112 Pakistan, 22,924 Rest of World.

2.3 Known Data Quality Concern: Non-Response Bias

Only ~49% of all respondents reported compensation (a standard limitation of this survey). Within the Pakistan subgroup specifically, respondents who reported salary had a higher median experience (7 years) than those who did not (5.5 years) — suggesting salary reporters skew more senior. This means the final Pakistan sample likely overrepresents more experienced, higher-earning individuals relative to the true underlying respondent population. Directionally, this bias would tend to *understate* rather than overstate any gap favoring Rest-of-World, since it inflates Pakistan's observed compensation relative to its true distribution.

3. Unadjusted Findings

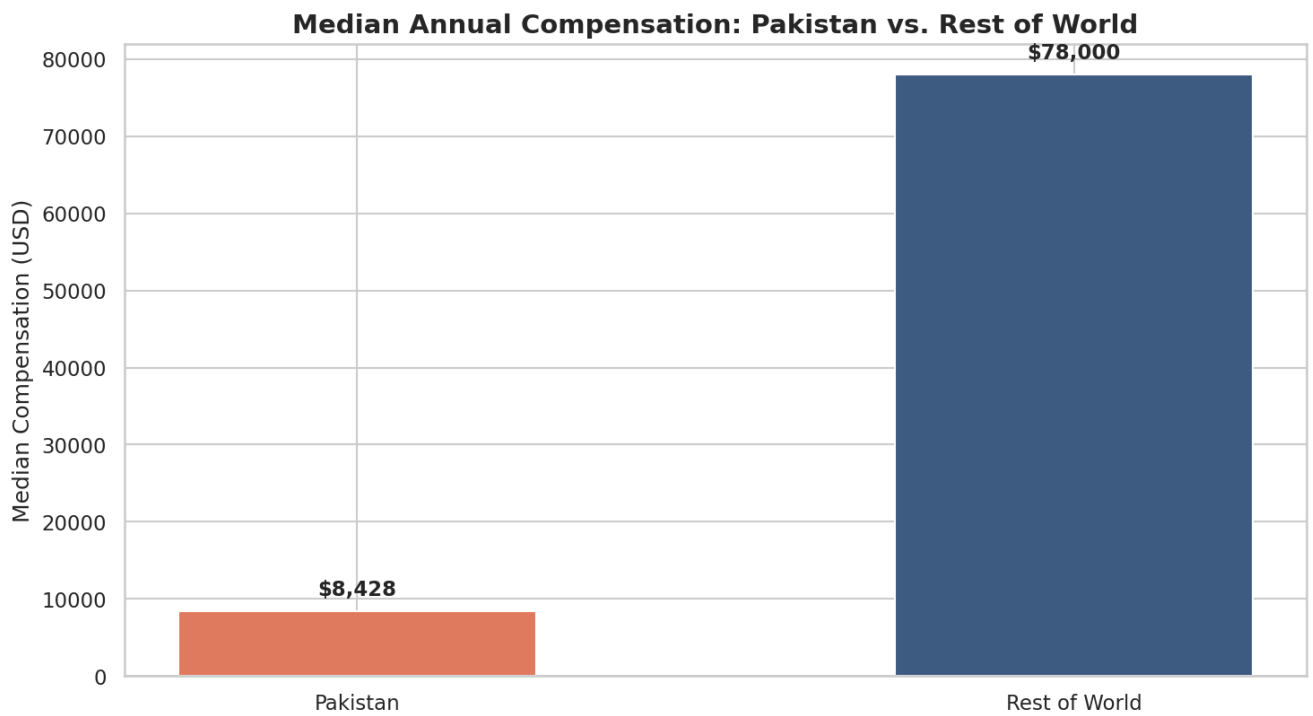
3.1 Descriptive Statistics

Group	n	Median	Mean	Std Dev
Pakistan	112	\$8,428	\$15,413	\$21,977
Rest of World	22,924	\$78,000	\$96,384	\$95,240

Both distributions are heavily right-skewed, confirmed via histogram, boxplot, and Q-Q plot inspection. Variances are highly unequal between groups, ruling out a standard Student's t-test.

3.2 Hypothesis Testing

Three tests were run to triangulate the result across different distributional assumptions:



Source: 2025 Stack Overflow Developer Survey (n=112 Pakistan, n=22,924 Rest of World)

Figure 1: Median compensation comparison

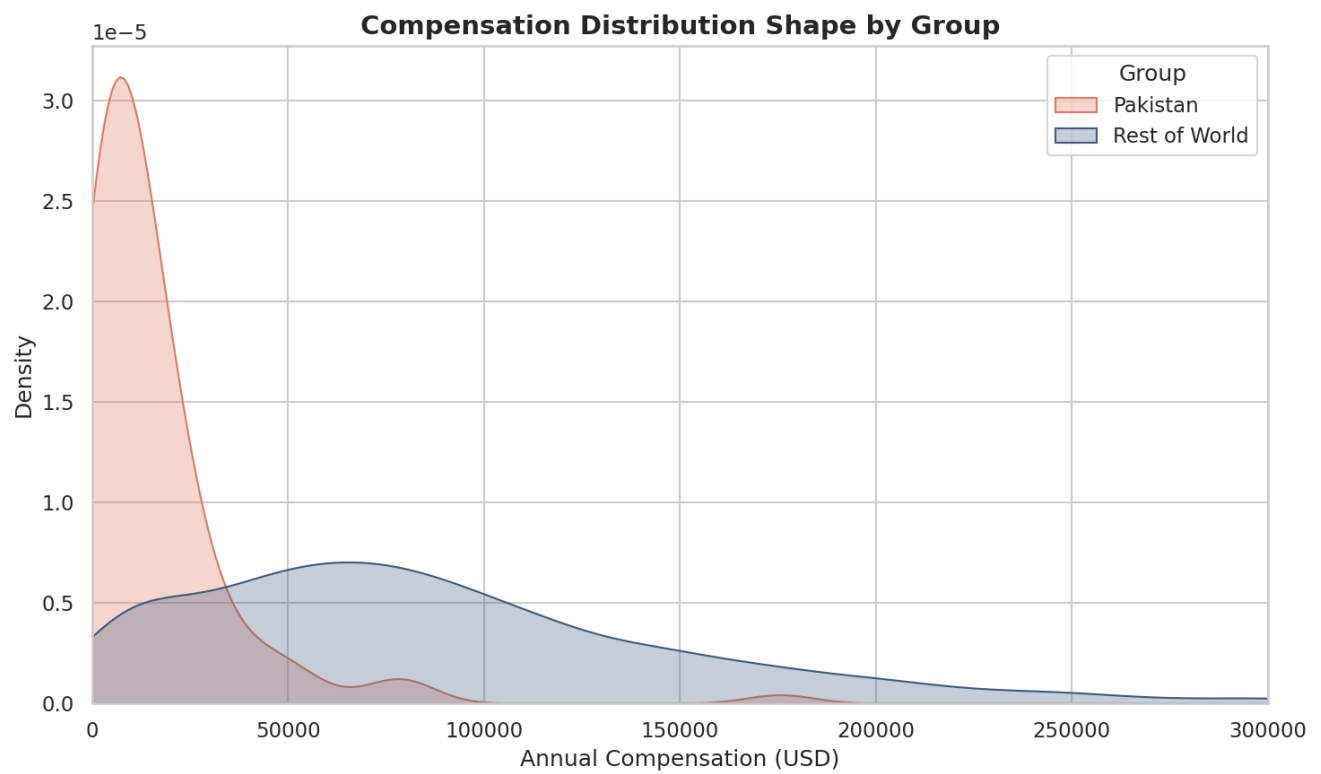


Figure 2: Compensation distribution shape by group

Test	Result
Welch's t-test (raw)	$t = -37.32, p < .001$
Welch's t-test (log-transformed)	$t = -19.09, p < .001$
Mann-Whitney U (non-parametric)	$p < .001$

All three agree: the unadjusted compensation difference is statistically significant.

3.3 Effect Size

Cohen's $d = 0.85$ (raw scale), **1.87 (log scale — considered more reliable** given raw-scale variance is inflated by extreme high earners in the Rest-of-World group). By conventional benchmarks (0.2/0.5/0.8 = small/medium/large), this is a large effect — not merely a statistically detectable one, but a substantively meaningful one.

Caution: with $n=22,924$ on one side, statistical significance alone was near-guaranteed even for a trivial difference. The large effect size, not the p-value, is what makes this finding worth reporting.

4. Confounder-Adjusted Findings

The unadjusted gap does not yet answer the core question — it doesn't account for the fact that Pakistan respondents are, on average, less experienced (median 7 years vs. 15 years for Rest-of-World). Two independent methods were used to test whether the gap survives this control.

4.1 Method 1: Stratified Comparison

Experience Band	Pakistan n	RoW n	Pakistan Median	RoW Median	Ratio
0–5 years	33	1,993	\$2,950	\$23,203	7.9x
5–10 years	50	5,215	\$10,861	\$51,071	4.7x
10–15 years	17	4,662	\$16,856	\$75,410	4.5x
15–25 years	7	6,107	\$25,284	\$92,812	3.7x*
25+ years	5	4,947	\$18,000	\$111,435	6.2x*

*Small Pakistan sample — directional only, not statistically reliable.

The gap persists in every single experience band without exception. The relative gap narrows somewhat from entry-level (~8x) to mid-career (~4.5x) before the small top-band samples make the pattern less certain.

4.2 Method 2: OLS Regression

Model: `log(ConvertedCompYearly) ~ YearsCode + CountryGroup`

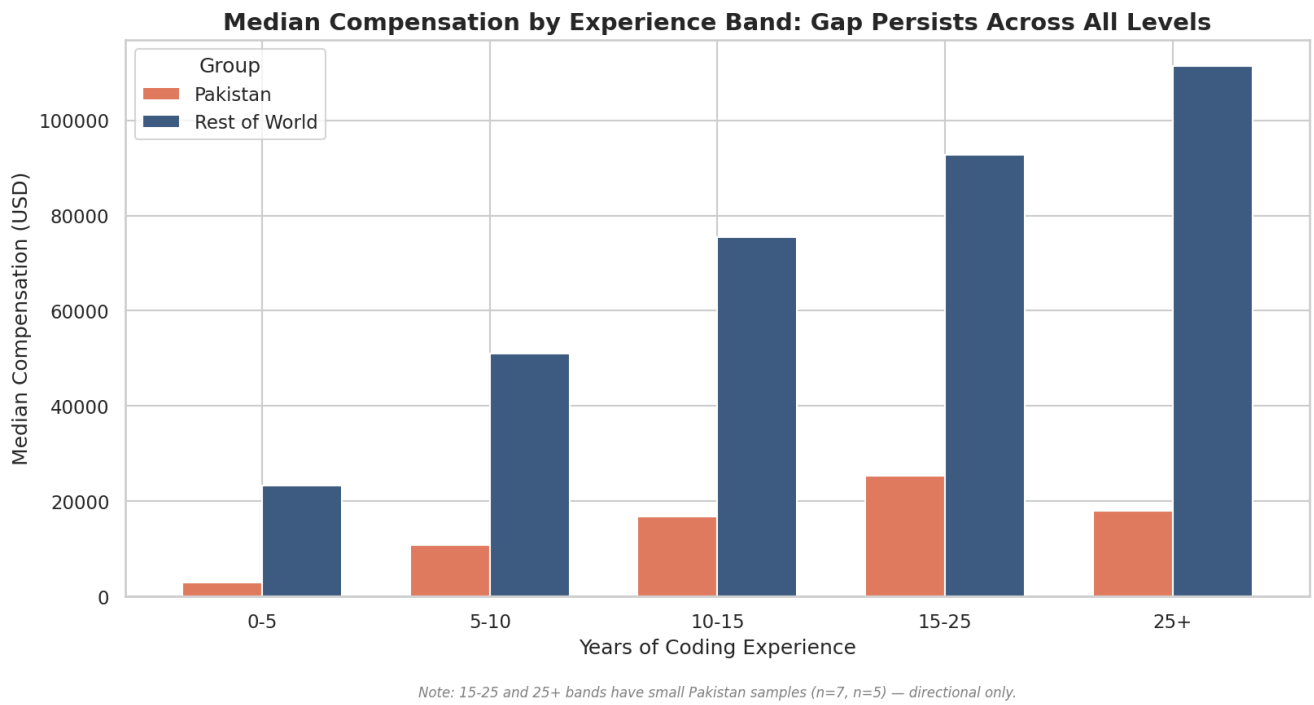


Figure 3: Compensation gap by experience band

Term	Coefficient	p-value
CountryGroup (Rest of World)	1.6713	< .001
YearsCode	0.0397	< .001

$R^2 = 0.167$. Result confirmed robust to heteroskedasticity-consistent (HC3) standard errors (coefficient and significance essentially unchanged).

Translated: holding years of experience constant, Rest-of-World respondents earn approximately **5.3x** what Pakistan respondents earn. This sits within the range produced by the stratified method (3.7x–7.9x across bands), providing convergent evidence from two independent approaches.

Each additional year of coding experience is associated with ~4.05% higher compensation, holding country group constant — confirming experience is a genuine driver, just not one that explains away the country-level gap.

Model limitation: $R^2 = 0.167$ means experience and country group together explain only ~17% of salary variance. The majority of variation is driven by factors not included in this model — see Section 6.

5. Business Insights

Insight	Confidence
The raw compensation gap is large and statistically robust	High — confirmed by 3 independent tests, large effect size
The gap persists after controlling for experience	High — confirmed by 2 independent methods (stratification, regression) converging on a consistent estimate
The gap narrows somewhat at mid-career before becoming uncertain at senior levels	Low-Moderate — senior bands have very small Pakistan samples (n=5–7)
Experience and country together explain only ~17% of salary variation	High — direct model statistic; other factors clearly matter

6. Recommendations

1. Treat this as a starting signal for further investigation, not a standalone conclusion.

Evidence: large, robust gap surviving experience control. Confidence: high that a gap exists, low on root cause. Assumption: nominal USD, not purchasing-power adjusted. Impact: prevents both premature over-claiming (discrimination) and under-reacting (dismissing a real signal). Risk: either extreme is a real risk if this analysis is over-interpreted.

****2.** Expand the model to include role/seniority, company size, and remote-work status before drawing stronger conclusions.****** Evidence: $R^2 = 0.167$ — most variance is currently unexplained. Confidence: high that these variables would materially change the estimate. Impact: necessary before this could inform

actual compensation policy.

3. Adjust for purchasing power parity (PPP) before using this for market benchmarking.

Evidence: nominal USD comparison conflates real pay differences with cost-of-living differences.

Confidence: high that PPP-adjustment would narrow the apparent gap substantially. Impact: prevents both underpaying local talent and overestimating true disparity.

7. Interpretation Caveats

- **Correlation/association, not causation.** This analysis rules out "experience alone explains the gap." It does not identify *why* the remaining gap exists, and does not establish discrimination or any specific causal mechanism.
 - **Nominal USD, not purchasing power.** A substantial nominal gap is expected between Pakistan and higher cost-of-living countries independent of any equity concern.
 - **Statistical significance $\neq$ practical significance, and vice versa.** Significance here was near-guaranteed by sample size; the large effect size is what makes the finding substantively meaningful.
 - **If a stakeholder claims this "proves" discrimination:** the correct response is that "not explained by experience" is a narrower, defensible claim than "caused by unfair treatment." Confirming the latter would require controlling for role, seniority, company size, and purchasing power at minimum — none of which were available in this dataset.
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8. Limitations & Future Work

- Small Pakistan sample size in senior experience bands ($n=5-7$) limits confidence in that portion of the findings
 - Salary non-response correlates with experience within the Pakistan subgroup, likely biasing the final sample toward more senior respondents than the true population
 - Self-selected survey population — findings apply to Stack Overflow respondents, not the tech workforce broadly, in Pakistan or elsewhere
 - Future iterations should incorporate `DevType` (role), `OrgSize` (company size), remote-work status, education level, and a purchasing-power-parity adjustment to better isolate the source of the remaining gap
 - Gender-based analysis (the project's original scope) was not possible due to the absence of a gender field in the 2025 survey's public release; this remains a potential avenue if a future survey year restores that field
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Appendix: Reproducibility

All analysis code, cleaning logic, and statistical outputs are available in the accompanying notebooks (`notebooks/01_data_cleaning.ipynb` through `04_confounder_analysis.ipynb`). See the repository root `README.md` for setup instructions.